Basic Programming 2

You think you can code?
This problem will test you on various basic programming techniques.
You are given two integers N and t; and then an array A of N integers (0-based indexing).
Based on the value of t, you will perform an action on A.

t	Action Needed
1	Print "`Yes`" if there are two integers $x \in A$ and $y \in A$ such that $x \neq y$ and $x + y = 7\,777$, or "`No`" otherwise (without the quotes)
2	Print "`Unique`" if all integers in A are different; or print "`Contains duplicate`" otherwise (without the quotes)
3	Find and print the integer that appears $> \frac{N}{2}$ times in A; or print -1 if such integer cannot be found
4	Find and print the median integer of A if N is odd; or print both median integers of A if N is even (separate them with a single space)
5	Print integers in A that fall between a range $[100 \ldots 999]$ in sorted order; (print a single space between two integers)

Input

The first line of the input contains an integer N and t ($3 \leq N \leq 200\,000$; $1 \leq t \leq 5$).
The second line of the input contains N non-negative 32-bit signed integers.

Output

For each test case, output the required answer based on the value of t.

Scoring

There are 20 hidden test cases that test various requirements of this problem.
All 20 test cases will be tested.
Each hidden test case worth 5 points (the 5 sample test cases below worth 0 point).

Sample Input 1

```
7 1
1 7770 3 4 5 6 7
```

Sample Output 1

```
Yes
```

Sample Input 2

```
7 2
1 2 3 4 5 6 7
```

Sample Output 2

```
Unique
```

Sample Input 3

```
7 3
1 1 1 1 2 2 2
```

Sample Output 3

```
1
```

Sample Input 4

```
8 4
8 1 4 3 6 7 5 2
```

Sample Output 4

```
4 5
```

Sample Input 5

```
7 5
210 999 1000 543 321 99 777
```

Sample Output 5

```
210 321 543 777 999
```